SIH 26248 — Research Brief & Design Decisions

For Claude Project: research, analysis, architecture, and reporting only

1. Critical instruction: do not build

This project is strictly for research, analysis, technical planning, and documentation. Do NOT build, develop, implement, or modify the actual software project. Do NOT generate a codebase, create or edit source files, execute commands, install packages, initialize repositories, or make changes to project files. Do not start implementation automatically, even after the architecture is finalized. Deliver a comprehensive research and engineering report, with architecture diagrams, workflows, comparison matrices, technical specifications, implementation roadmap, test strategy, risks, and traceability. Pseudocode may be used only when useful to explain a design; do not turn it into a working application. Continue research and documentation only unless I explicitly authorize implementation in a separate instruction.

2. Project objective

Develop a rigorous, evidence-backed proposal for SIH 2026 Problem Statement 26248: “Immersive Multi-Domain Decision-Making Trainer for Degraded Communication Environments.” The intended direction is a production-oriented blueprint for a simulator that could be evaluated for future institutional military training, subject to official requirements, security review, procurement, and user validation. Do not claim that a prototype is production-ready or officially approved.

3. Source files to read first

- problem_statement_26248.pdf — official problem statement.
- military_wargame_solutions_decision_as...pdf — collected research on existing wargame and decision-making solutions.
- SIH_2026_PPT_One_Page_Checklist.pdf — presentation checklist.
- The user also states that they have the official SIH 2026 PowerPoint template. It must be verified and used as the presentation-format authority.

First verify that each file is accessible and readable. Do not claim to have analyzed inaccessible files. If necessary, ask the user to upload them directly. Extract the exact requirements and constraints from the source documents before adding external research. Clearly label source-derived facts, external research, engineering judgments, assumptions, and proposals.

4. Answers to the 18 open questions

Q-01 — Handwritten terms and quantities

A platoon is understood by the user as a small group of soldiers. “12 km” means twelve kilometers, but what it measures is not known. “20 TP” may mean 20 troops, but this is uncertain. The user does not remember what “50 tracks” means. “SAE” and “FOV” remain unidentified. Do not convert these into firm requirements; research context and ask for clarification only if needed.

Q-02 — Military symbology reference

The user supplied this preferred visual reference: https://www.rff.com/documentation/military-symbols-stencil.pdf. Treat it as a preferred reference, not proof of an official Indian Army standard. Research whether it is suitable, what standard or symbol conventions it reflects, its licensing/use terms, and whether a documented standard is more appropriate.

Q-03 — Participants and roles

The simulator should support a team of trainees. Claude should research and recommend appropriate roles rather than assuming a command structure or exact team size.

Q-04 — Intended setting

Prioritize a military classroom or training institution, with an instructor supervising a team of trainees. Field use is not an initial requirement.

Q-05 — Immersion and interface

Prioritize a high-quality web application with interactive maps, dashboards, timelines, and military symbols. Research whether selective 3D features would materially improve training, but do not make VR headsets a core dependency.

Q-06 — Hardware and connectivity

Dedicated-GPU computers are available for the demo. Internet is normally available. However, the final architecture should also be evaluated for offline local-network operation; demo internet availability does not imply the deployed system may depend on external services.

Q-07 — Domains

Use an integrated multi-domain approach: land, air, electronic warfare (EW), and cyber should influence the evolving scenario, information available to trainees, and communications. Research and scope the interactions so they are credible and feasible; do not create four disconnected mini-games.

Q-08 — Offline operation

Support internet-connected demonstrations and offline operation over a local network without internet. Core simulation, scenario execution, maps, user roles, and session recording should not depend on external cloud services. Identify and justify any exceptions. Local-network operation is not automatically air-gapped or security-approved.

Q-09 — Team and skills

Team size: 6. Available skills include React, Java, Python, game development, databases, networking, DevOps, and UI/UX or visualization. Experience is mixed, with some stronger developers. Recommend a production-oriented stack based on technical merit, then distinguish ideal long-term architecture from feasible SIH implementation.

Q-10 — Terrain

The user prefers real-world terrain and geographic data. Research data sources, licensing, offline packaging, performance, security, and any applicable mapping constraints. Do not assume that sensitive or restricted datasets are available.

Q-11 — Interface language

English-only professional interface. Use consistent terminology, readable map labels, clear status indicators, and a professional military-oriented visual language.

Q-12 — After-action review (AAR)

Preferred: interactive replay, detailed PDF report, and exportable structured data. Include evidence-backed requirements and instructor review. Research established AAR practices and define measurable indicators without treating automated scores as objective truth.

Q-13 — SIH presentation template

The user states they have the official SIH 2026 PowerPoint template. Verify and use the actual template before designing slides. Treat the template and the supplied checklist as constraints; do not substitute generic slide advice.

Q-14 — Current stage and deadline

Current stage: researching and preparing the solution idea. The user knows the deadline and will provide it separately. Do not invent or assume a deadline. Prioritize research, architecture, and presentation preparation before implementation.

Q-15 — Algorithm and error analysis

Cover three distinct layers: software correctness (bugs, incorrect calculations, simulation inconsistencies, failed events); trainee decision analysis (decisions, delays, information verification, consequences); and information uncertainty (effects of incomplete, contradictory, delayed, or misleading reports). Define defensible indicators for each.

Q-16 — Assessment

Use transparent, evidence-backed indicators and an explicit rubric; the instructor makes the final judgment. Avoid unsupported claims about decision quality. Identify the validation or subject-matter-expert input needed for formal assessment.

Q-17 — Opposing side and event control

Use a hybrid approach: scripted scenario events plus automated opposing behavior, with optional instructor control or a human opposing team. Compare realism, repeatability, implementation effort, and training value; recommend what should be implemented first.

Q-18 — Interoperability

Build an independent simulator. Integration with existing military simulators is outside the initial scope. Do not assume access to proprietary military systems. Modular interfaces may be considered for future extensibility, but do not let integration requirements inflate the initial scope.

5. Research scope and required analysis

Requirements extraction

Read all three PDFs completely. Extract explicit functional and non-functional requirements, constraints, deliverables, evaluation criteria actually stated, ambiguities, and dependencies. Do not invent judging criteria.

Independent research

Research current public information about relevant military wargaming and decision trainers, serious games, multi-domain simulation, uncertainty and misinformation training, degraded communications, offline-first systems, real-time geospatial visualization, scenario authoring, instructor controls, assessment, AAR, and deployment security.

Existing-solution comparison

For each relevant system, compare verified purpose, architecture, technology stack, decision mechanics, misinformation/uncertainty, communication degradation, multi-user design, visualization, instructor tooling, assessment, AAR, deployment, offline operation, interoperability, strengths, limits, and relevance. Distinguish production systems, prototypes, academic work, and marketing claims.

Technology decisions

Compare credible alternatives for frontend and visualization, simulation engine, backend, data layer, real-time communications, event processing, offline operation, AI/ML, identity/access control, observability, and deployment. Evaluate licensing, maturity, maintenance, hardware, security, performance, and integration complexity. Recommend one coherent stack, not an unfiltered list.

Architecture

Provide system context, component architecture, module boundaries, data model, event/message lifecycle, scenario lifecycle, communication-degradation model, uncertainty model, multi-user synchronization, instructor workflows, AAR/replay, security boundaries, offline behavior, deployment topology, and observability.

AI/ML

For each proposed model, specify responsibility, necessity, inputs/outputs, data requirements, latency/hardware, offline behavior, metrics, failure modes, fallback, explainability, and human oversight. Prefer deterministic rules or conventional software where more reliable and testable.

Roadmap and verification

Give phased implementation steps, prerequisites, deliverables, owners/roles, dependencies, effort estimates with assumptions, risks, tests, and definitions of done. Distinguish SIH demo scope from future production engineering and approval.

PPT strategy

Use the actual official template and checklist. For every required slide, give its message, evidence, architecture/visual, measurable value, and claims needing validation. No fabricated results, fake screenshots, unsupported accuracy figures, or claims of official adoption.

Traceability

Create a matrix linking official requirement → relevant PPT requirement → user idea → design decision → module → task → test → evidence → status. Maintain assumptions, unresolved questions, decisions, risks, dependencies, rejected alternatives, and a final acceptance audit.

6. Evidence and research rules

- Use current web research and verify source currency.
- Prefer primary sources: official documentation, standards bodies, government/defence publications, peer-reviewed papers, project documentation, and source repositories.
- For significant findings, record the claim, direct URL, publication/update date if available, access date, evidence type, what the source proves, relevance, and limitations.
- Do not fabricate citations, URLs, benchmarks, specifications, adoption claims, or military preferences.
- Distinguish sourced facts, engineering judgments, assumptions, and proposals.
- Do not assume that all military agencies share one preferred technology stack. Separate verified defence-sector use from general industry adoption and recommendation.

- Do not assume access to classified information, proprietary systems, restricted networks, or non-public datasets.
- Favor testability, reproducibility, modularity, security, maintainability, accessibility, observability, and a clearly defined need for each technology.
- The long-term ambition is production-oriented quality, but do not call the SIH prototype production-ready or approved for official use.

7. Initial response expected from Claude

1. Confirm access to the three PDFs and inventory their contents.
2. Read the official problem statement and PPT checklist completely; extract exact requirements and constraints.
3. Read the existing-solutions PDF and inventory every identifiable system or reference.
4. Identify ambiguities, missing information, and research questions.
5. Propose a structured research plan.
6. Begin evidence-based research and comparison; do not jump straight to a final tech stack.
7. Present cited findings, comparison matrices, provisional conclusions, and unresolved questions.
8. Prepare the architecture, technology decision matrix, build roadmap, test strategy, and PPT strategy for review only after research is sufficiently complete.
9. Stop at the architecture approval gate. Do not generate implementation code.

8. Open items and assumptions to keep visible

- Clarify SAE, FOV, “50 tracks,” what 12 km measures, and whether “20 TP” means 20 troops.
- Research the status and suitability of the RFF military-symbols stencil reference; do not assume it is an official Indian Army standard.
- Research and recommend exact participant count and role structure.
- Confirm whether any official mentor/sponsor has additional constraints or acceptance criteria.
- Verify terrain data sources and licensing before selection.
- Verify the official SIH PPT template and exact submission requirements.
- The user will provide the deadline separately.
- Research appropriate AAR export formats; preferred outputs are interactive replay, PDF, and structured data.
- Research defensible trainee assessment indicators and what expert validation is required.
- Research whether any public interoperability standards are worth keeping as future extension points; integrations are not in initial scope.

9. Approval boundary

The immediate deliverable is a comprehensive, cited research and engineering report and a proposed blueprint. No coding or implementation is authorized. Even after the architecture is reviewed, continue documentation only unless the user explicitly authorizes implementation in a separate instruction.